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Efficacy of repeated low-level red-light therapy for myopia management in children and adolescents: a systematic review and meta-analysis

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Abstract

Background To systematically evaluate the efficacy of repeated low-level red-light (RLRL) therapy, compared to single-vision spectacles (SVS), for controlling myopia progression in children and adolescents.

Methods A comprehensive search of electronic databases, including PubMed, Embase, Cochrane Library, was conducted up to Sept 30, 2025 according to the inclusion and exclusion criteria. The change of axial length (AL), spherical equivalent refraction (SER) and subfoveal choroidal thickness (SFCT) were main outcomes. Pooled mean differences (MD) with 95% confidence interval (CI) were calculated and heterogeneity was quantified using I^2.

Results Fourteen randomized controlled trials involving 1,942 participants were included. The mean difference in AL change favored RLRL at 1 month (MD: -0.07 mm, 95% CI: -0.09 to -0.06, $P < 0.00001$), 3 months (MD: -0.15 mm, 95% CI: -0.18 to -0.11, $P < 0.00001$), 6 months (MD: -0.22 mm, 95% CI: -0.26 to -0.17, $P < 0.00001$), and 12 months (MD: -0.33 mm, 95% CI: -0.39 to -0.27, $P < 0.00001$). Similarly, RLRL significantly slowed SER progression at 3 months (MD: 0.17 D, 95% CI: 0.13 to 0.21, $P < 0.00001$), 6 months (MD: 0.38 D, 95% CI: 0.26 to 0.49, $P < 0.00001$), and 12 months (MD: 0.67 D, 95% CI: 0.53 to 0.82, $P < 0.00001$). Furthermore, RLRL induced a significant increase in SFCT at all time points (16.72 µm at 1 month; 19.46 µm at 3 months; 28.21 µm at 6 months; 30.86 µm at 12 months; all $P < 0.00001$). Heterogeneity was substantial for AL and SER outcomes but the treatment effect was consistently favorable.

Conclusion In conclusion, this meta-analysis suggests that RLRL therapy maybe an effective intervention for slowing myopia progression in children and adolescents in the short to medium term. Based on current evidence, it appears to be a promising and generally well-tolerated treatment option. However, long-term safety, sustained efficacy, and optimal treatment protocols require further confirmation through extended follow-up studies.

Trial registration The trial was registered with the International Prospective Register of Systematic Reviews (PROSPERO; registration ID: CRD420251139693).

Keywords Myopia, Repeated low-level red-light, Axial length, Spherical equivalent refraction, Subfoveal choroidal thickness

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Background

Myopia, commonly known as nearsightedness, has emerged as a global public health crisis in recent decades, characterized by excessive axial elongation of the eyeball and resulting in blurred distance vision [1, 2]. Characterized by excessive axial elongation, myopia has shown an alarming increase in prevalence across all geographic regions, particularly in East and Southeast Asia [3]. Current projections indicate that nearly 5 billion people worldwide will be myopic by 2050, with approximately 1 billion expected to develop high myopia [3]. This trend represents not merely a refractive inconvenience but a substantial healthcare challenge, as high myopia constitutes a leading cause of visual impairment and irreversible blindness due to its associated sight-threatening complications, including myopic maculopathy, retinal detachment, open-angle glaucoma, and premature cataract formation [2]. The pathophysiology of myopia progression involves complex interactions between genetic predisposition and environmental factors, with the modern lifestyle characterized by increased near-work activities and reduced time spent outdoors identified as significant risk factors [4]. The standard approach to managing myopia has evolved dramatically from simply correcting refractive error with single-vision spectacle (SVS) to actively intervening to slow myopia progression, recognizing that axial elongation increases the risk of future pathological complications [5, 6].

Current mainstream interventions for myopia control include pharmacological approaches, primarily low-dose atropine, and optical strategies such as myopia-control spectacles, orthokeratology and specialized multifocal contact lenses. While these modalities have demonstrated efficacy, each presents limitations that affect their widespread applicability. Low-dose atropine (0.01%, 0.025% and 0.05%) has demonstrated efficacy in slowing myopia progression along a concentration-dependent response [7]. However, side effects like photophobia and impaired near vision, though milder than with higher concentrations, persist, and a concerning "rebound effect" upon cessation has been observed [8], though this may be mitigated by employing a tapering schedule at the end of treatment. Optical interventions face challenges related to compliance, accessibility, cost, and potential complications such as microbial keratitis associated with overnight lens wear.

Despite these advances, limitations in safety, accessibility, cost, and variable individual response underscore the need for alternative, non-invasive, and convenient treatment options. It is within this context that repeated low-level red-light (RLRL) therapy has emerged as a groundbreaking and promising intervention. In recent years, RLRL therapy, employing a phototherapy device emitting delivering specific wavelengths of around 650 nm and a power density (below 0.023mW/mm^2) well below the thermal damage threshold, has emerged as a novel and promising intervention for myopia control [9]. The treatment protocol commonly requires children are required to gaze at a fixed observation port for 3 min, twice daily with an interval of at least 4 h between sessions and a minimum frequency of 5 days per week. The clinical study for RLRL has grown rapidly and axial length (AL), spherical equivalent refraction (SER), as well as subfoveal choroidal thickness (SFCT) are all important indicators for evaluating the efficacy of RLRL. A landmark randomized controlled trial (RCT) [10] demonstrated remarkable efficacy, with children in the RLRL group showing significantly less AL elongation and SER progression at 12 months compared to those wearing SVS. Some cohort studies have largely confirmed these findings, with AL and SER effectively controlled compared to the control group [11, 12]. This has led to the widespread adoption of RLRL, particularly in China.

However, the evidence base is not without heterogeneity and controversy. Results vary across studies, potentially due to differences in device parameters (brand, wavelength, power), patient demographics, and baseline myopia severity. Therefore, while individual studies are promising, a comprehensive and quantitative synthesis of the available evidence is urgently needed. A meta-analysis can explore sources of heterogeneity and provide a more precise estimate of the treatment effect. Two previous meta-analysis included few RCTs [13, 14], and another one [15] lacked 1-month and 3-months follow-up results. This systematic review and meta-analysis aims to synthesize the current latest evidence from RCTs to definitively evaluate the efficacy of RLRL therapy, compared to single-vision spectacles, for controlling myopia progression in children and adolescents.

Materials and methods

This meta-analysis was conducted and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [16]. The protocol of this meta-analysis was registered with the International Prospective Register of Systematic Reviews (PROSPERO; registration ID: CRD420251139693; available from https://www.crd.york.ac.uk/PROSPERO/view/CRD420251139693) prospectively. The filled PRISMA 2020 checklist is provided in Supplementary Table S1.

Search strategy

A comprehensive literature search was performed and the detailed study selection process according to PRISMA 2020 guidelines [17]. The electronic databases searched included PubMed, Embase, Cochrane Library, *ClinicalTrials.gov*. Only studies published in English were included. The search strategy was designed

to incorporate a combination of Medical Subject Headings (MeSH) terms and free-text keywords related to the core concepts: "myopia", "low-level red-light", and " randomized controlled trial". The search string for PubMed, which was adapted for other databases, was: ((myopia) OR (myopias) OR (nearsightedness) OR (nearsightednesses)) AND ((repeated low-level red-light) OR (low-level red-light) OR (RLRL) OR (phototherapy) OR (laser therapy)) AND ((randomized clinical trial) OR (RCT)). No restrictions were placed on the publication year or language. Additionally, the reference lists of all included studies and relevant review articles were manually screened. Furthermore, to minimize publication bias, we searched pre-print servers (medRxiv, bioRxiv) for relevant unpublished or recently presented work. These are collectively referred to as "other sources" in the study selection flow diagram (Fig. 1).

Eligibility criteria and study selection

The study selection process followed the PICOS (Population, Intervention, Comparator, Outcomes, Study Design) criteria. The population of interest was children and adolescents with myopia (defined as spherical equivalent refraction ≤ -0.50 D). The intervention was RLRL therapy, either as a standalone treatment or in combination with SVS. The comparator was SVS. No other myopia intervention was provided in the control group. The outcomes were between-group differences in changes of AL (mm), SER (diopter, D), SFCT (μm) with minimum 1-month follow up. Study designs were RCTs published as articles in English. Studies were excluded if they were reviews, case reports, conference abstracts without full data, involved animal subjects, enrolled adult populations or included participants with secondary myopia due to other ocular or systemic syndromes.

The study selection was conducted independently by two reviewers. After removing duplicates, they first screened the titles and abstracts of all retrieved records against the eligibility criteria. The full texts of potentially relevant articles were then obtained and assessed in detail. Any disagreements between two reviewers at either stage were resolved through discussion or by consulting a third reviewer.

Data extraction and quality assessment

Data from the included studies were extracted independently by two reviewers using a standardized, pre-piloted data extraction form. Any discrepancies were judged by the third reviewer and resolved by consensus. The extracted data included: (1) Study characteristics: first author, publication year, study location, study design, sample size, and follow-up duration; (2) Participant characteristics: mean age, baseline AL, SER and SFCT; (3) Intervention details: RLRL device parameters (wavelength, power), and treatment regimen (session duration, frequency); (4) Comparator details; and (5) Outcome

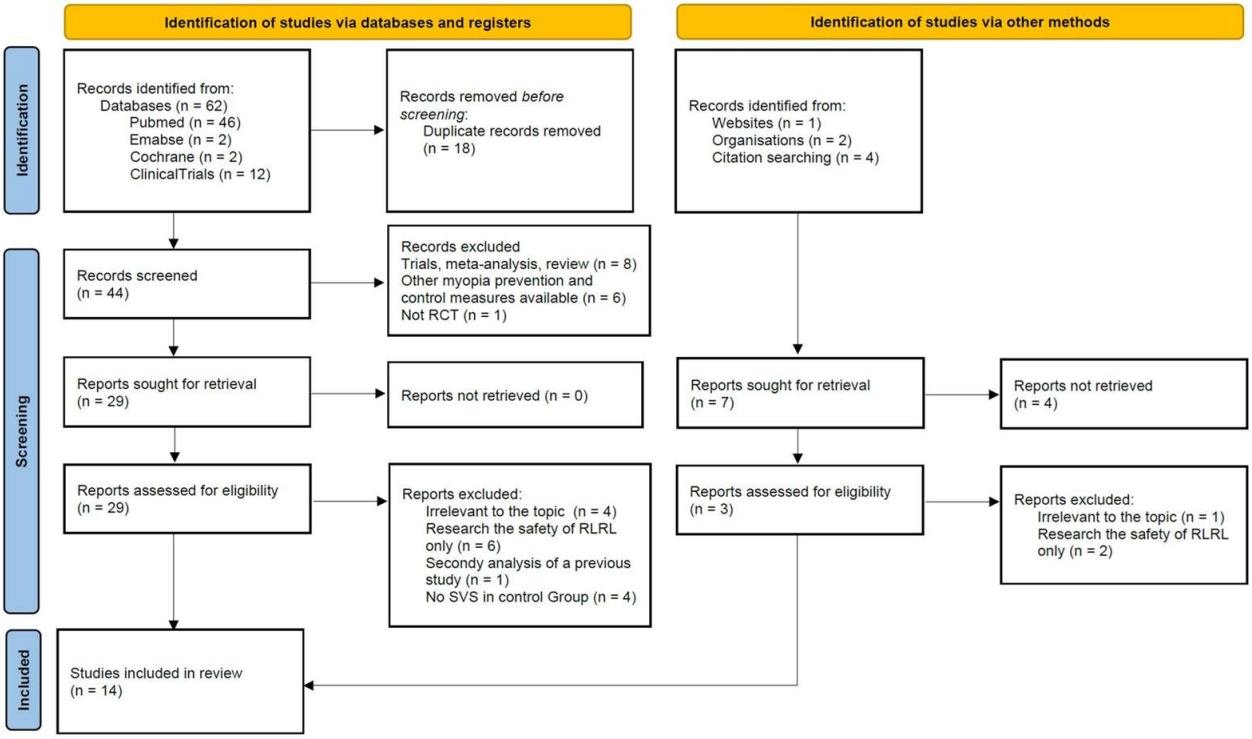


Fig. 1 Flow diagram of study selection process. The number of studies at each level (n) is shown

data: changes of AL, SER and SFCT from baseline to follow-up for each group. All continuous variables were reported as mean ± standard deviation (SD); when presented as interquartile ranges or 95% confidence intervals (95% CI), these data should be converted appropriately to be presented as mean ± SD [18]. If the necessary data were not reported in the published article, we attempted to contact the corresponding authors to request the missing information. Studies with critically missing outcome data that could not be retrieved or reliably estimated were excluded from the quantitative synthesis for that specific outcome. Furthermore, AL was considered as the primary outcome and secondary outcomes included SER and SFCT.

Risk of bias assessments of eligible studies

The methodological quality and risk of bias of each included RCT were assessed independently by two reviewers using the revised Cochrane Risk of Bias tool for randomized trials (RoB 2) [19]. The tool evaluates five domains: (D1) bias arising from the randomization process; (D2) bias due to deviations from intended interventions (assessed for the 'effect of assignment to intervention'); (D3) bias due to missing outcome data; (D4) bias in measurement of the outcome; and (D5) bias in selection of the reported result. Each domain was judged as 'Low risk', 'Some concerns', or 'High risk' based on a series of questions. A detailed assessment with justifications for each domain judgement in all included RCTs is provided.

Statistical analyses

All statistical analyses were performed using Review Manager (RevMan) software (version 5.4.1). For changes in AL, SER and SFCT, the treatment effect was expressed as the mean difference (MD) with a 95% CI by therapy duration (1 month, 3 months, 6 months and 12 months). Statistical heterogeneity among the included studies was assessed using Chi-square tests. If no heterogeneity ($P > 0.05$ or $I < 50\%$) was observed, the fixed effect model was applied to analyse data. If heterogeneity was observed, the random effect model was used and a meta-regression analysis was performed to identify the possible sources of variability. Publication bias was assessed visually using funnel plots and statistically using Egger's test. A P-value of less than 0.05 was considered statistically significant for the treatment effect.

Results

Results of the literature search

The systematic literature search initially identified 62 records from databases and 7 additional records from other sources. After removing duplicates, 44 unique records were screened by title and abstract. Of these, 29 full-text articles were assessed for eligibility. Following a detailed evaluation, 11 articles were excluded with reasons: four were irrelevant to the topic, six investigated only the safety of RLRL without efficacy data, four also included no SVS in the control group and one was a secondary analysis of a previously included study. Consequently, 14 randomized controlled trials (RCTs) [10, 20–32] met the inclusion criteria and were included in the quantitative synthesis. The study selection process is detailed in the PRISMA flow diagram (Fig. 1).

Basic characteristics of the included studies and risk of bias assessment

The 14 included RCTs enrolled a total of 1,942 participants, with sample sizes ranging from 34 to 336. The baseline characteristics of these studies are summarized in Table 1. The participants' mean age ranged from approximately 8.51 to 11.88 years, covering a pediatric and adolescent population. The studies encompassed a spectrum of myopia severity, with baseline spherical SER ranging from – 1.13 D to -7.75 D and AL from 23.80 mm to 26.50 mm. The follow-up duration for the outcomes varied from 1 to 12 months.

The assessment of the methodological quality of the included RCTs, performed using the Cochrane RoB 2 tool [19], is presented in Fig. 2. And a detailed assessment with justifications for each domain judgement is provided in Supplementary Table S2. The majority of studies demonstrated a low risk of bias in the domains of random sequence generation and incomplete outcome data, indicating robust randomization procedures and adequate handling of participant attrition. However, a significant proportion of studies had a high risk of bias in the domains of blinding of participants and personnel (performance bias), which is a common challenge in device intervention trials. Selective reporting bias was generally judged as low across the studies.

Axial length

The meta-analysis of the effect of RLRL on AL is presented in Fig. 3. The pooled results demonstrated a consistent and statistically significant reduction in AL in the RLRL group compared to the control group across all time points and analyses. At 1 month (Fig. 3A), a significant reduction in AL was already observed in the RLRL group (MD: -0.07 mm, 95% CI: -0.09 to -0.06, $P < 0.00001$) but with substantial heterogeneity ($P = 0.008$, $I^2 = 63\%$). At 3 months (Fig. 3B), the treatment effect became more pronounced, with the MD increasing to -0.15 mm (95% CI: -0.18 to -0.11, $P < 0.00001$). Heterogeneity remained high ($P < 0.00001$, $I^2 = 83\%$). At 6 months (Fig. 3C), the analysis confirmed a significant treatment effect, with a mean difference MD of -0.22 mm (95% CI: -0.26 to -0.17, $P < 0.00001$). The heterogeneity was considerable

Table 1 Study characteristics of the 14 randomized controlled trials enrolled in this meta-analysis

Study (First author, year)	Treatment	Number, *n*	Gender (Male/Female)	Age, years	Baseline AL, mm	Baseline SER, D	Baseline SFCT, μm	Follow-up duration, month
Guihua Liu, 2025 [20]	RLRL	130	61/69	10.7 ± 1.7	26.50 ± 1.03	-7.75 ± 1.91	249 ± 69	1, 3, 6, 12
	SVS	68	30/38	10.5 ± 1.6	26.38 ± 0.99	-7.65 ± 1.70	261 ± 64	
Guihua Liu, 2024 [21]	RLRL	32	16/16	9.37 ± 1.69	24.71 ± 0.92	-2.91 ± 1.27	NA	6, 12
	SVS	36	17/19	9.55 ± 1.13	24.58 ± 0.64	-2.61 ± 0.98	NA	
Hongyan Chen, 2023 [22]	RLRL	46	27/19	9.00 ± 1.90	24.62 ± 0.97	-2.54 ± 1.04	259.00 ± 51.46	6, 12
	SVS	40	25/15	8.98 ± 1.92	24.57 ± 0.76	-2.29 ± 0.77	273.08 ± 54.37	
Jing Dong, 2023 [23]	RLRL	56	26/30	10.3 ± 2.07	24.7 ± 1.04	-3.13 ± 1.91	NA	6
	Sham device	55	30/25	9.86 ± 1.41	24.6 ± 0.96	-2.82 ± 1.86	NA	
Kai Cao, 2024 [24]	RLRL	168	86/82	9.1 ± 2.0	23.9 ± 1.1	-1.4 ± 1.6	291.4 ± 71.6	12
	SVS	168	90/78	9.0 ± 1.9	23.8 ± 1.0	-1.3 ± 1.5	299.3 ± 68.2	
Ke Yang, 2025 [25]	RLRL	26	10/16	9.00 (8.00, 10.00)	24.30 ± 0.87	-1.13 (-1.38, -1.00)	NA	12
	SVS	26	13/13	9.00 (8.00, 10.00)	23.93 ± 0.66	-1.13 (-1.25, -1.00)	NA	
Lei Tian, 2022 [26]	RLRL	91	NA	9.66 ± 1.65	24.31 ± 0.92	-2 (-3.25, -1.25)	290.5 (242, 352.5)	6
	SVS	88	NA	9.47 ± 1.59	24.20 ± 0.85	-2 (-2.75, -1.25)	296 (244, 352)	
Nelofar Deen 2025 [27]	RLRL	17	10/7	11.61 ± 1.55	24.24 ± 0.69	-2.26 ± 0.89	NA	1, 3
	SVS	17	10/7	11.88 ± 1.41	24.54 ± 0.76	-2.16 ± 1.04	NA	
Ruili Xiong, 2023 [28]	RLRL	60	28/32	10.52 ± 1.53	24.52 ± 0.68	-2.30 ± 0.85	215.31 ± 58.52	1, 3, 6, 12
	SVS	60	27/33	10.37 ± 1.61	24.66 ± 0.89	-2.58 ± 1.15	215.50 ± 57.71	
Wen Zhou, 2024 [29]	RLRL-0.37mW	43	27/16	8.51 ± 1.51	24.24 ± 0.81	-1.79 ± 0.96	255.09 ± 55.76	1, 3, 6
	RLRL-0.60mW	47	25/22	8.77 ± 1.43	24.11 ± 0.89	-2.05 ± 0.88	248.11 ± 44.49	
	RLRL-1.20mW	44	24/20	8.68 ± 1.39	24.38 ± 0.90	-2.10 ± 1.36	257.84 ± 63.46	
	SVS	43	26/17	8.83 ± 1.53	24.44 ± 0.93	-2.09 ± 0.90	267.58 ± 54.86	
Yan Xu, 2024 [30]	RLRL	97	63/34	10.4 ± 2.4	25.93 ± 1.03	-5.88 ± 1.69	NA	6, 12
	SVS	95	44/51	11.2 ± 2.1	25.72 ± 0.83	-5.75 ± 1.17	NA	
Yinghui Xiong, 2024 [31]	RLRL	36	17/19	8.83 ± 2.06	24.38 ± 0.87	-2.47 ± 1.39	251.83 ± 65.27	1, 3, 6
	SVS	37	22/15	9.00 ± 2.00	24.47 ± 0.58	-2.22 ± 0.72	274.76 ± 63.79	
Yu Jiang, 2022 [10]	RLRL	117	NA	10.4 (8.0, 13.0)	24.54 ± 0.67	-2.49 ± 0.92	NA	1, 3, 6, 12
	SVS	129	NA	10.5 (8.1,13.0)	24.62 ± 0.86	-2.67 ± 1.06	NA	
Zhaoxin Jiang, 2025 [32]	RLRL	35	19/16	11.32 ± 2.05	24.59 ± 0.77	-2.37 ± 0.69	244.79 ± 53.11	6, 12
	SVS	35	19/16	11.37 ± 2.08	24.58 ± 0.82	-2.21 ± 0.71	244.94 ± 45.25	

Data are presented as mean ± standard deviation for normally distributed continuous variables, otherwise data are expressed as median and interquartile range. RLRL, repeated low-level red-light; SVS, single-vision spectacle; AL, axial length; SER, spherical equivalent refraction; SFCT, subfoveal choroidal thickness. NA, not applicable

($P < 0.00001$, I^2 = 89%). At 12 months (Fig. 3D), the strongest effect was observed, with a mean difference of -0.33 mm (95% CI: -0.39 to -0.27, $P < 0.00001$) in favor of the RLRL group. Heterogeneity was also high at this time point ($P < 0.00001$, I^2 = 89%).

Spherical equivalent refraction

The results for SER progression, detailed in Fig. 4, corroborated the findings for AL, indicating a significant slowing of myopia progression with RLRL treatment over a longer duration. At 1 month (Fig. 4A), the MD in SER change was 0.03 D (95% CI: -0.02 to 0.08). This positive point estimate indicated a slight treatment benefit, but the result was not statistically significant ($P = 0.27$). At 3 months (Fig. 4B), the pooled results indicated that RLRL could slow SER progression significantly more than control group (MD: 0.17 D, 95% CI: 0.13 to 0.21, $P < 0.00001$) without substantial heterogeneity ($P = 0.15$, $I^2 = 37\%$). At 6 months (Fig. 4C), the analysis confirmed a significant treatment effect, with the MD increasing to 0.38 D (95% CI: 0.26 to 0.49, $P < 0.00001$), and substantial heterogeneity was present ($P < 0.00001$, I^2 = 82%). At 12 months (Fig. 4D), the most substantial effect on SER was observed (MD: 0.67 D, 95% CI: 0.53 to 0.82, $P < 0.00001$) with substantial heterogeneity ($P < 0.00001$, $I^2 = 88\%$).

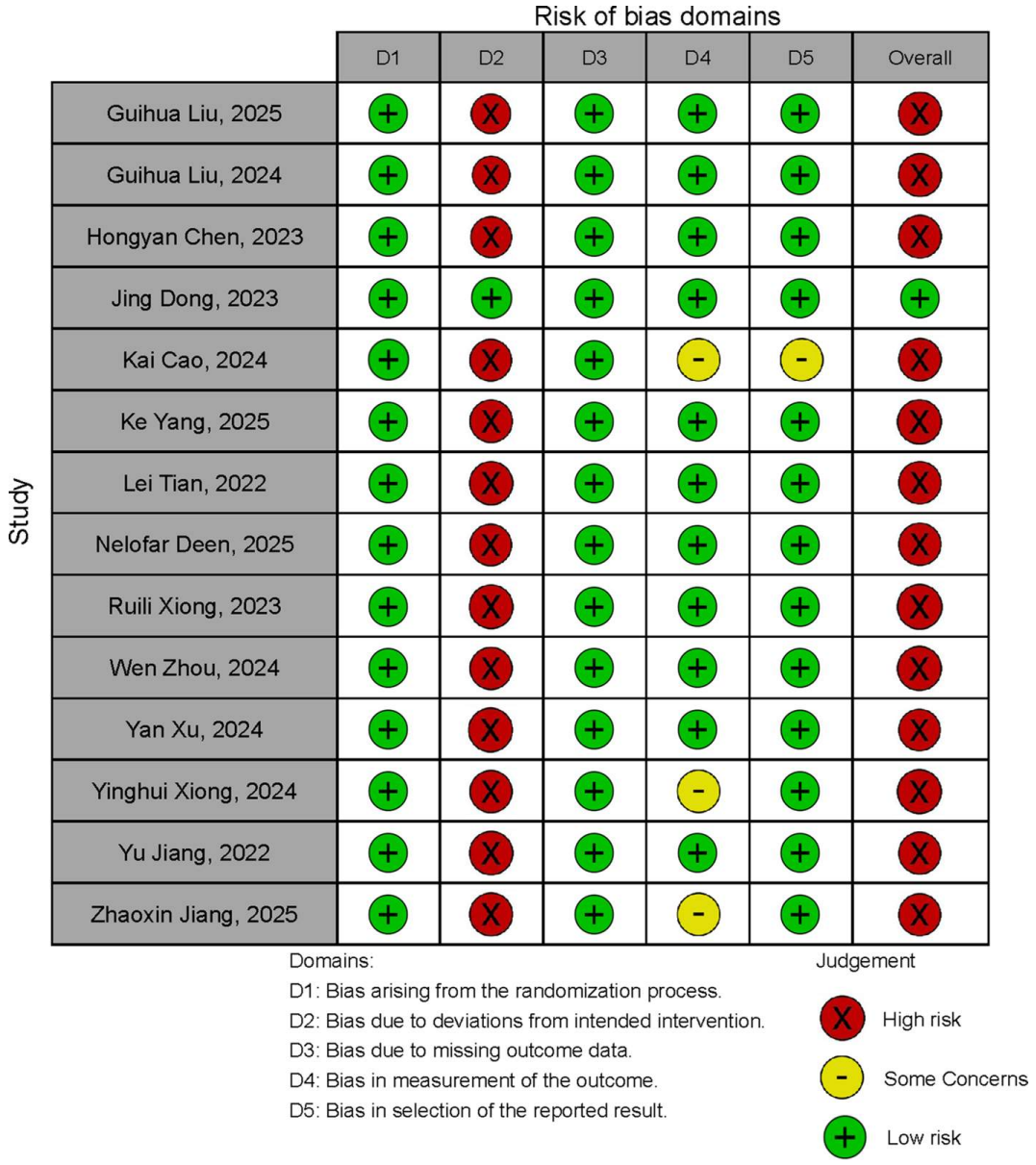


Fig. 2 Summary of bias risk for all included studies. (**A**) Risk of bias graph. (**B**) Risk of bias summary. Bias risk was classified as low (+), unclear (?), or high (-)

Subfoveal choroidal thickness

Subfoveal choroidal thickness represented the anatomic change after myopia control. The analysis results of SFCT change were presented in Fig. 5. At 1 month, the pooled analysis of five studies demonstrated a significant increase in SFCT in the RLRL group compared to the control group at the 1-month follow-up, with a MD of 16.72 μm (95% CI: 12.79 to 20.66, $P<0.00001$). The analysis used a fixed-effects model due to low heterogeneity ($P=0.65$, $I^2=0\%$). At 3 months, the SFCT remained significantly greater in the RLRL group (MD: 19.46 μm, 95% CI: 10.26 to 28.66, $P<0.00001$) with substantial heterogeneity ($P=0.03$, I^2 = 72%). At 6 months, the analysis confirmed a significant treatment effect, with a pooled mean difference of 28.21 μm (95% CI: 19.49 to 36.92, $P<0.00001$) favoring RLRL. High heterogeneity was observed ($P<0.0001$, I^2 = 78%,), and a random-effects model was applied. The 12-month analysis included 5 studies and there was considerable heterogeneity among the studies ($P<0.00001$, I^2 = 89%). Despite this variability, the pooled result indicated that RLRL treatment led to a significantly greater increase in SFCT compared to the control, with a mean difference of 30.86 μm (95% CI: 20.55 to 41.18, $P<0.00001$).

Publication bias and meta-regression

Funnel plots and Egger's test were employed to assess potential publication bias. The funnel plot showed an

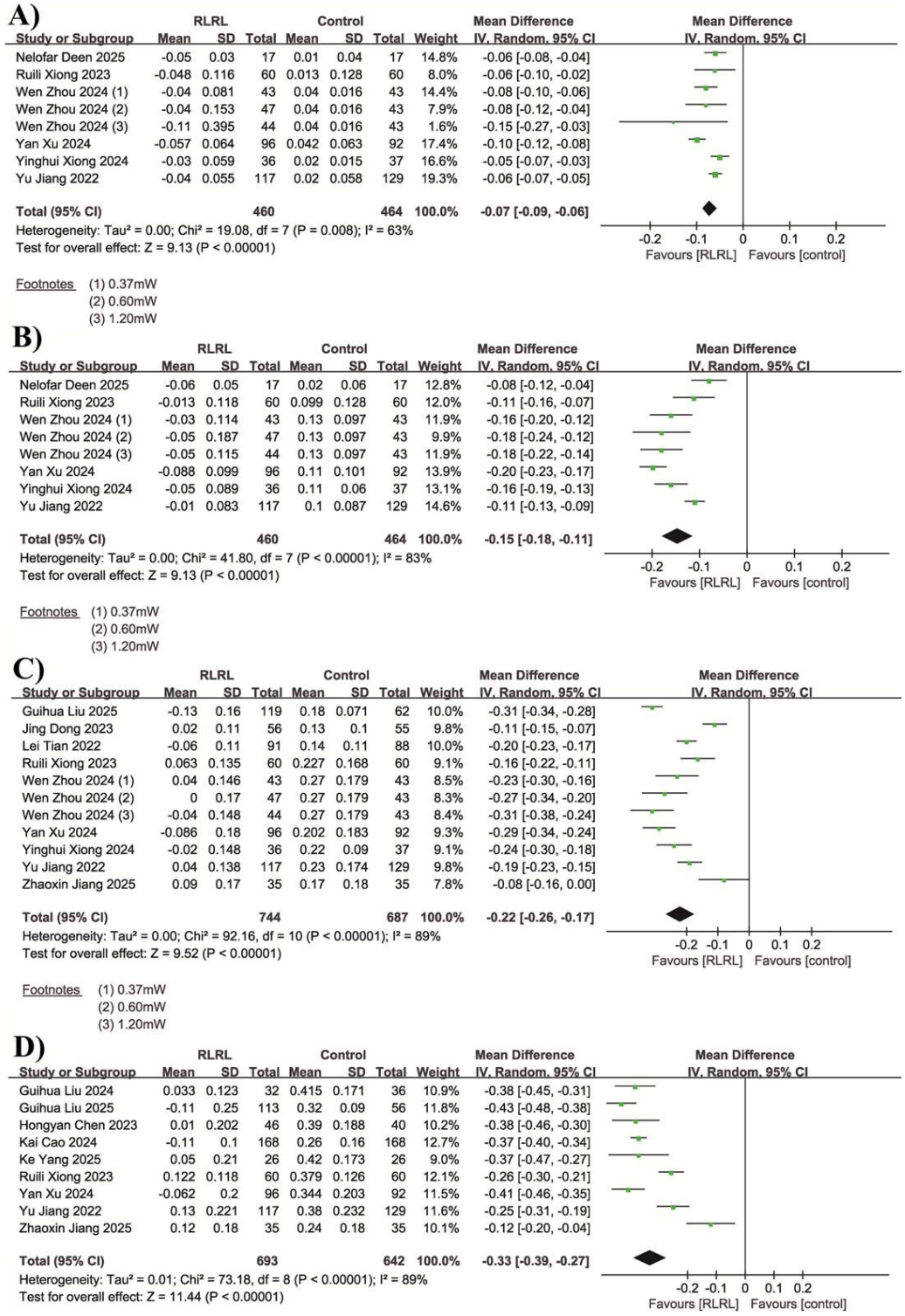


Fig. 3 Forest plots comparing the change in AL between the RLRL group and the control group at different follow-up time points. Results are presented as mean difference with 95% confidence interval. **A**) 1-month results. **B**) 3-month results. **C**) 6-month results. **D**) 12-month results. RLRL, repeated low-level red-light

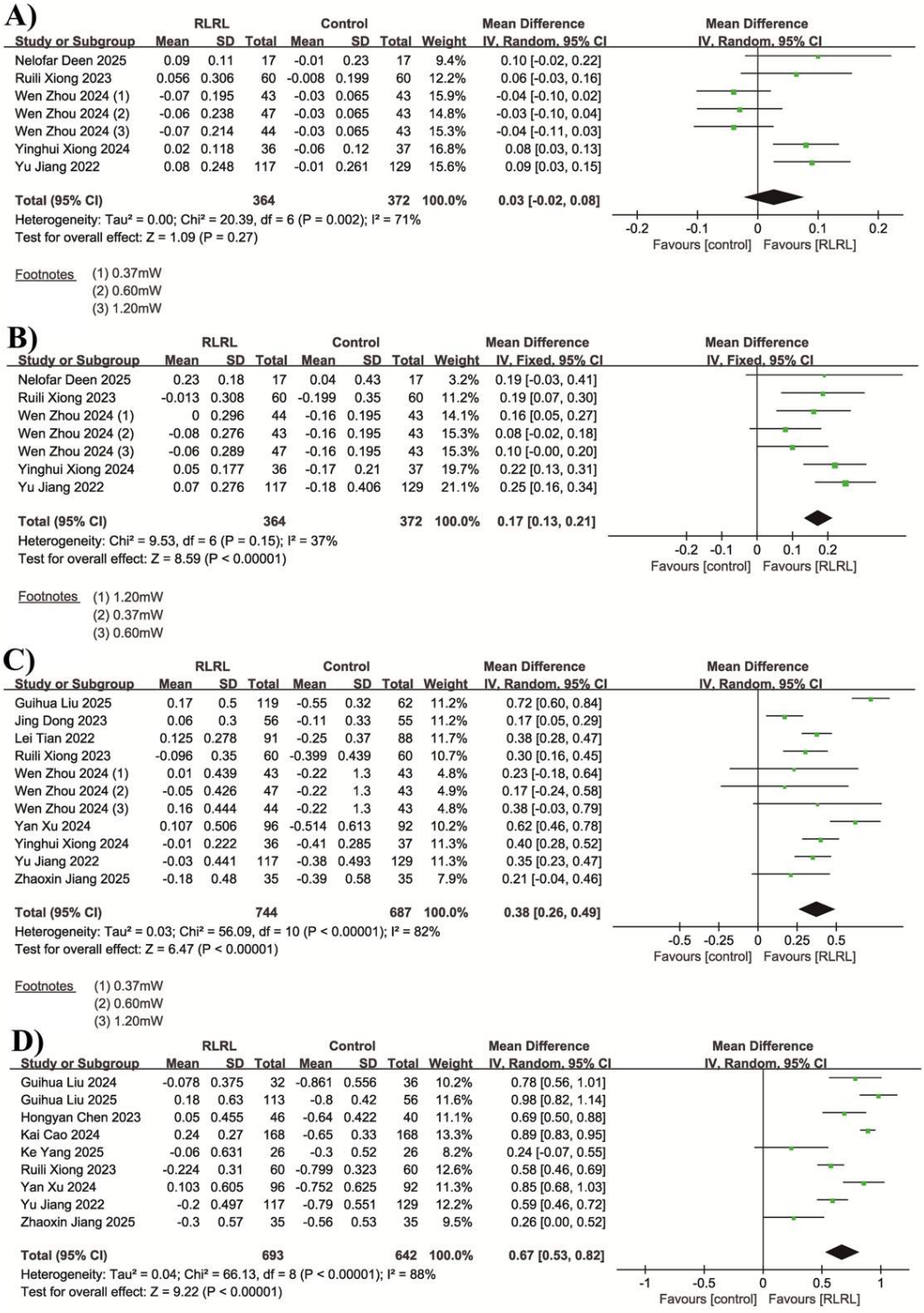


Fig. 4 Forest plots comparing the change in SER between the RLRL group and the control group at different follow-up time points. Results are presented as mean difference with 95% confidence interval. **A**) 1-month results. **B**) 3-month results. **C**) 6-month results. **D**) 12-month results. RLRL, repeated low-level red-light

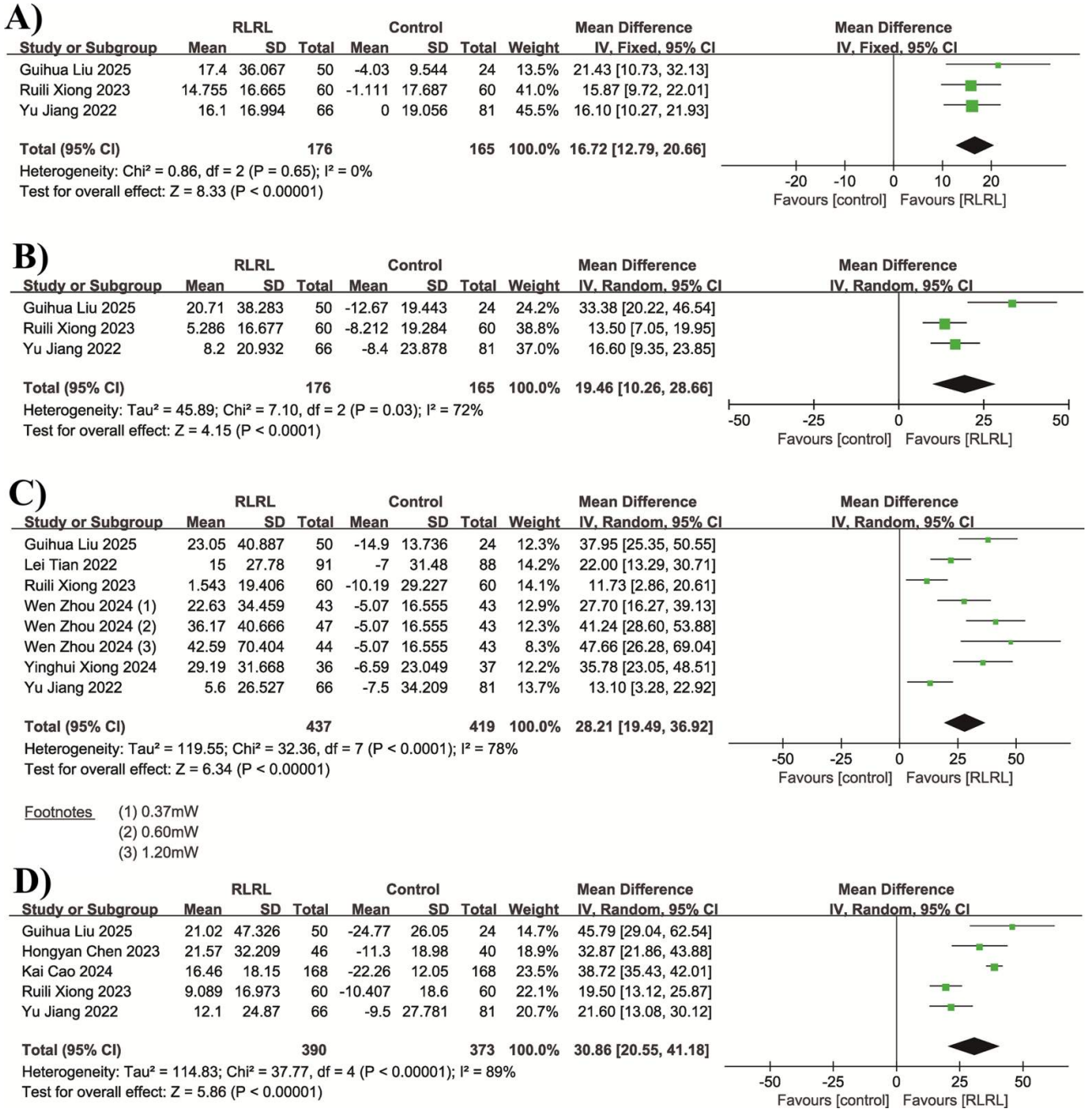


Fig. 5 Forest plots comparing the change in SFCT between the RLRL group and the control group at different follow-up time points. Results are presented as mean difference with 95% confidence interval. **A**) 1-month results. **B**) 3-month results. **C**) 6-month results. **D**) 12-month results. RLRL, repeated low-level red-light

asymmetric distribution (Supplementary Figure S1). In addition, Egger's ($p < 0.001$) revealed significant publication bias.

In the meta-regression results, the heterogeneity was explained by RLRL device power ($P = 0.008$), but the heterogeneity was not explained by the other factors.

Discussion

This systematic review and meta-analysis of 14 randomized controlled trials provides a quantitative synthesis of the current evidence, indicating that RLRL therapy may be an effective intervention for slowing myopia progression in children and adolescents. The primary findings suggest that RLRL is associated with a reduction in axial elongation and a slowing of SER progression compared to SVS, alongside a concurrent increase in SFCT.

The consistency of these trends across multiple studies, despite considerable clinical heterogeneity, positions RLRL as a promising approach within the evolving strategies to address the global myopia epidemic [33].

The most compelling evidence from our analysis pertains to the control of axial length (AL), the paramount anatomical risk factor for future myopic pathology. The pooled data revealed a progressive and substantial treatment effect. The mean difference in AL change was – 0.07 mm at 1 months, which intensified to - 0.33 mm at 12 months. Epidemiological studies suggest that each additional millimeter of axial elongation exponentially increases the lifetime risk of sight-threatening complications such as myopic maculopathy, retinal detachment, and glaucoma [5, 34]. Therefore, an intervention capable of attenuating growth by 0.20 mm annually represents a clinically profound effect, potentially altering the lifelong visual prognosis for a child. This effect was even more pronounced in the high myopia, suggesting that RLRL may be particularly crucial for this most at-risk population, potentially mitigating their trajectory towards visual impairment [35].

The findings for SER progression powerfully corroborate the AL data, yet they reveal an informative temporal dynamic. The effect at 1 month was negligible and non-significant (MD = 0.03 D), but became clear and statistically significant by 3 months (MD = 0.17 D), and continued to strengthen at 6 months (MD = 0.38 D) and 12 months (MD = 0.67 D). This discrepancy between the early (1-month) and later SER results is critical to interpret. It may indicate that the full refractive consequence of RLRL is cumulative and becomes more discernible over a longer duration. Alternatively, it might reflect an initial physiological adjustment period where the eye's refractive components are stabilizing. This temporal dissociation underscores a critical principle in myopia control trials: axial length serves as a more sensitive, objective, and early indicator of therapeutic efficacy than refraction alone. The structural change (AL) manifests and is detectable earlier, while the full refractive consequence becomes apparent later, reinforcing the primacy of AL as the primary outcome measure in clinical trials and long-term management [36, 37].

Beyond the compelling efficacy data for AL and SER, our analysis provides a crucial, albeit preliminary, mechanistic insight through the outcome of subfoveal choroidal thickness (SFCT). The pooled results demonstrated a highly significant and sustained increase in SFCT in the RLRL group at all time points, from + 16.72 μm at 1 month to + 30.86 μm at 12 months. This finding is not merely a statistical association; it provides a plausible anatomical correlate for the treatment's mode of action. The choroid is a highly dynamic, vascular structure increasingly recognized as a key player in emmetropization [38]. A thicker choroid is consistently associated with slowed ocular growth and emmetropic shifts [39]. The choroidal thickening following RLRL therapy maybe a rapid vascular response to the retinal signaling changes [40]. Scleral hypoxia is a proposed stimulus for myopic elongation, driven by increased metabolic demand from the RPE [41]. By enhancing choroidal perfusion and potentially oxygen tension, RLRL may alleviate this hypoxic drive, thereby stabilizing the scleral extracellular matrix and inhibiting the expression of matrix metalloproteinases that lead to its remodeling and thinning [42].

While establishing efficacy is paramount, a comprehensive appraisal of RLRL must equally weigh its safety profile. The included RCTs, along with dedicated safety studies, consistently report that short-term RLRL use is generally well-tolerated. The most commonly reported adverse events are transient and mild, including glare, photophobia, and eye fatigue, with no study reporting vision-threatening complications within their trial periods. These findings support the short-term clinical acceptability of the intervention. The most significant limitation of the current safety evidence is the absence of long-term data. All included trials have follow-up durations of 12 months or less, with only a handful of extensions to 24 months. The safety of RLRL over many years or decades of use, through childhood and into adulthood, remains unknown. Potential delayed effects on retinal pigment epithelium function, choroidal vasculature, or lens transparency cannot be ruled out without prolonged surveillance. It is critical to distinguish between short-term tolerability and long-term safety. The reported adverse event profile supports good short-term tolerability. However, the absence of long-term data, particularly regarding the potential cumulative effects on the retina and choroid, represents a significant uncertainty and a key gap in the current evidence base.

A notable feature of our analysis was the substantial statistical heterogeneity for the AL and SER outcomes at later time points. While this heterogeneity complicates the precise estimation of a single effect size, it is not unexpected and, importantly, offers valuable clinical insights. Our meta-regression analysis identified device power as a significant source of this heterogeneity. The included trials utilized RLRL devices with different power outputs [29], suggesting a potential dose-response relationship, with higher power potentially yielding a greater treatment effect. This variability in device parameters across studies is a major contributor to heterogeneity and a critical area for future standardization. This confirms that the treatment effect is not uniform but is context-dependent, particularly on the technical specifications of the intervention.

This meta-analysis has several limitations, many of which reflect the current state of the evidence. First, the

follow-up duration is limited. The efficacy reported at 12 month should not be assumed to continue linearly beyond this time point, because the critical questions of sustained efficacy over many years, the existence of a "rebound effect" upon cessation, and the ultimate impact on pathologic myopia incidence remain unanswered. Second, the generalizability of the findings is currently skewed, as the vast majority of high-quality evidence comes from East Asian populations. This geographic and ethnic concentration represents a significant constraint on the external validity of our findings. The efficacy and safety of RLRL in other ethnic groups, with potentially different genetic predispositions and environmental exposures, require further validation. Third, the inability to perfectly blind participants and personnel to the intervention is an inherent limitation in device trials, introducing a potential for performance bias. Fourth, adherence to the twice-daily treatment protocol is a crucial practical determinant of efficacy, yet most included trials did not systematically report quantified compliance data. The lack of standardized compliance reporting limits our understanding of the real-world effectiveness and the dose-response relationship of RLRL therapy. This limits our understanding of the relationship between adherence level and treatment outcome and is a key area for standardization in future RLRL trials. Finally, the high heterogeneity, while informative, means that the pooled effect estimate should be interpreted as an average across varying contexts rather than a precise prediction for every individual.

Beyond the biological and methodological considerations discussed above, several practical limitations must be acknowledged for RLRL therapy. First, long-term adherence to a twice-daily, in-home treatment regimen can be challenging for children and families, and suboptimal compliance would directly undermine its efficacy. Second, the upfront cost of the RLRL device and the potential lack of insurance coverage may present a significant financial barrier, limiting access for many families. Third, the availability of RLRL devices, along with the necessary professional guidance for their use and monitoring, is currently not uniform globally, which restricts its widespread adoption. These factors of adherence, cost, and access are critical determinants of the real-world effectiveness and equity of RLRL as a myopia control strategy.

Conclusions

This meta-analysis of existing randomized controlled trials suggests that repeated low-level red-light (RLRL) therapy may be a promising intervention for slowing axial elongation and refractive progression in children and adolescents over a period of up to 12 months, compared to single-vision spectacles. The available evidence indicates a generally favorable short-term safety profile within the controlled settings of the included studies. However, these findings must be interpreted with caution due to several key limitations. The high degree of statistical heterogeneity across studies implies that the treatment effect is likely context-dependent, influenced by factors such as device parameters and patient characteristics, rather than universally uniform. Importantly, the current evidence base is predominantly short- to medium-term. Critical questions regarding long-term efficacy, safety over many years, the potential for a rebound effect upon cessation, and the optimization of treatment protocols remain unanswered. Therefore, while RLRL represents a novel and potentially valuable approach to myopia management, its role in clinical practice requires further clarification.

Abbreviations

RLRL	Repeated low-level red-light
SVS	Single-vision spectacles
AL	Axial length
SER	Spherical equivalent refraction
SFCT	Subfoveal choroidal thickness
MD	Mean differences
CI	Confidence interval
RCT	Randomized controlled trial
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses

Supplementary Information

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Supplementary Material 1

Supplementary Material 2

Supplementary Material 3: Supplementary Figure S1. Funnel plots for the assessment of potential publication bias. (A) Axial length. (B) Spherical equivalent refraction change. (C) Spherical equivalent refraction change. SE, standard error; MD, mean difference

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Author contributions

D.L and J.W had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Conception and design: D.L and J.W. Data collection: D.L, T.Y and J.W. Analysis and interpretation: D.L, G.M and J.W. Drafting the manuscript: D.L and J.W. Critical revision of the manuscript: D.L, T.Y and J.W. Supervision: J.W. All authors reviewed and approved the manuscript.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

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The authors declare no competing interests.

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